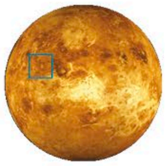
APHRODITE TERRA

Mead Crater

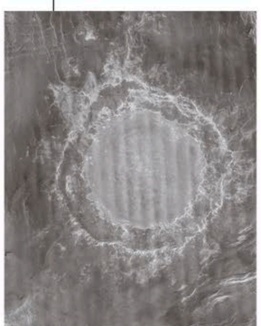


TYPE Multiringed crater

AGE Under 500 million years

DIAMETER 168 miles (270 km)

Mead is the largest impact crater on Venus—although compared to craters on the Moon and Mercury, it is not very large. Mead is a multiple-ringed crater whose inner ring is the rim of the crater basin. This encloses a smooth, flat floor, which hides a possible central peak. The crater floor was flooded at the time of impact as a result of impact melt or by volcanic lava being released from below the surface. This explains why a crater of Mead's size is so shallow; there is a drop of only about 0.6 miles (1 km) between the crater rim and the crater center.



LARGEST CRATER
Mead has two distinct rings. Ejecta lies between them and beyond the outer ring. The vertical bands running through the picture are a result of image processing.

LAVINIA PLANITIA

Saskia Crater

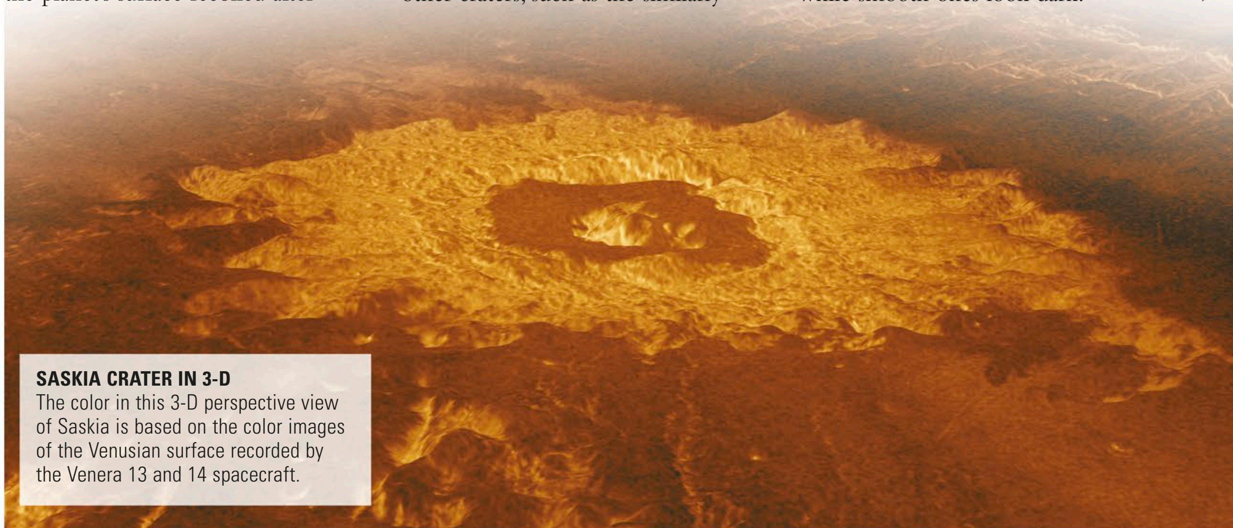
TYPE Central-peak crater

AGE Under 500 million years

DIAMETER 23 miles (37.1 km)

Saskia is a middle-sized crater, and its ejecta pattern is typical for its size. The ejecta blanket extends all the way around the crater's basin, suggesting that the impacting body smashed into the surface at a high angle. The crater has central peaks, formed as the planet's surface recoiled after being pushed down by the energy released during the impact. The original crater rim has collapsed and formed terraced walls. The incoming object must have been about 1.6 miles (2.5 km) across to produce a crater of this size. Images of Saskia and other craters, such as the similarly sized Danilova (30.3 miles/48.8 km wide) and Aglaonice (39.6 miles/63.7 km wide), which lie within a few hundred miles of Saskia, have been produced from radar data collected by Magellan. Raw radar images (such as the one above) do not show features as they would appear to the naked eye. Instead, brightness varies according to the smoothness of the surface—rough areas appear light, while smooth ones look dark.

THREE CRATERS
Saskia lies at the lower left of this 300-mile (500-km) wide segment of Lavinia Planitia. Above it are the Danilova and Aglaonice craters.



SASKIA CRATER IN 3-D
The color in this 3-D perspective view of Saskia is based on the color images of the Venusian surface recorded by the Venera 13 and 14 spacecraft.

LAVINIA PLANITIA

Stein Crater Field

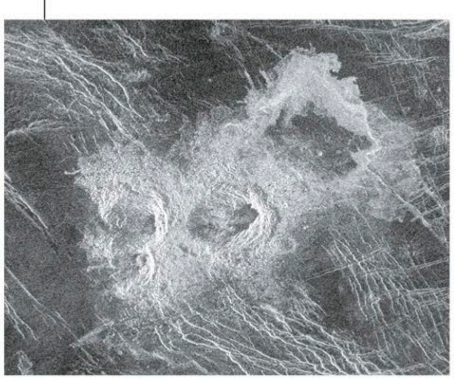
TYPE Crater field

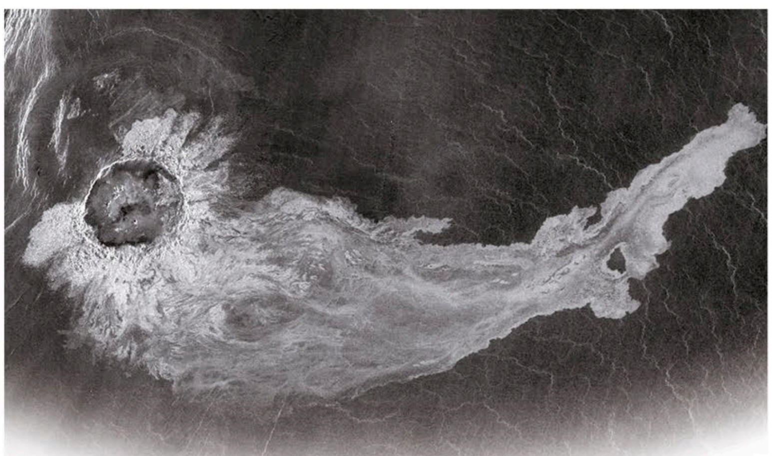
AGE Under 500 million years

DIAMETER 8.7 miles (14 km), 6.8 miles (11 km), and 5.6 miles (9 km)

Small asteroids heading for Venus's surface can be broken up by the planet's dense atmosphere. The resulting fragments continue heading toward the surface, striking it simultaneously within a relatively small area to form a crater field. The Stein field consists of three small craters. The two smallest ones overlap. Material ejected by all three craters extends mainly to the northeast, suggesting that the fragments struck from the southwest. Material melted by the impacts has formed flow deposits, also lying to the northeast.

STEIN TRIPLETS





CRATER AND OUTFLOW
A 372-mile (600-km) long, radar-bright flow of once-molten debris stretches to the east of Addams Crater.

AINO PLANITIA

Addams Crater

TYPE Central-peak crater

AGE Under 500 million years

DIAMETER 54 miles (87 km)

The large, circular Addams Crater measures almost 55 miles (90 km) across, but it is its long tail that makes this crater unusual. An asteroid hit the ground from the northwest and created a crater basin with an ejecta blanket stretching out beyond about three-quarters of the crater rim. Additionally, impact-melt ejecta and lava extend out from about a third of the rim, creating a mermaid-style tail to the east. The molten material flowed downhill for about 370 miles (600 km) from the impact site. The Magellan spacecraft found this area to be radar bright—that is, it bounced back a large portion of the radio waves that Magellan transmitted to it, which suggests it has a rugged surface. Venus's high surface temperature of about 867°F (464°C) allows ejecta to remain molten for a longer time than if it were on Earth. However, once the material cools below about 1,800°F (1,000°C), it becomes so viscous, it stops flowing. The crater is named after the American social reformer Jane Addams.

LADA TERRA

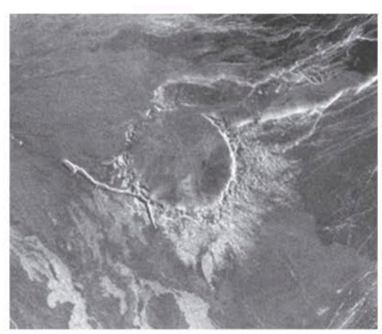
Alcott Crater

TYPE Degraded crater

AGE Under 500 million years

DIAMETER 41 miles (66 km)

Alcott is one of the few craters on Venus that have been modified by volcanic activity not associated with the crater's production. Many craters have floors flooded with lava that came up through the surface as the crater basin was formed. In Alcott's case, lava erupted elsewhere and then flowed over the crater. About half of the crater's rim is still visible, along with ejecta from the original impact lying to the south and east. A channel where lava once flowed touches the southwest edge of the crater.



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